

Comprehensive list of output variables

Variable types include instantaneous (I) values at the archival time step, means between archival times (M), and miNimum (N) or maXimum (X) values between archival times. For variable types M, N and X, the values of the time dimension correspond to the center of the time interval over which the minimum, maximum or average is calculated. For instance, the precipitation flux (pr) is given at times 0:30, 1:30, 2:30, etc., which correspond to the average precipitation flux over the time intervals 0:00–1:00, 1:00–2:00, 2:00–3:00, etc., respectively. The latter interval bounds are provided via the variable “time_bnds” present in the time series of variables of type M, N or X. Daily and monthly means are calculated from hourly and 3-hourly output. For some variables, only monthly means are archived.

For 3D atmospheric variables, data is provided on 11 pressure levels : 1000, 975, 950, 925, 900, 850, 800, 700, 600, 500 and 250 hPa. For 3D soil variables, data is provided at the 17 soil levels down to a depth of 15 m. However, variables related to soil moisture should be used with caution below a depth of 2 m, because deep-soil moisture cannot be considered to be in equilibrium in this ensemble.

The status column shows the availability of the variables via THREDDS: available at time of writing (T), available later (L), or not available (blank). When a variable is shared on THREDDS at a coarser frequency than what was originally saved, the THREDDS frequency is indicated in parenthesis.

Variable	Description	Frequency	Realm	Type	Status
baresoilFrac	Bare Soil Fraction	3-hr	Land 2D	I	
cape	Convective Available Potential Energy	3-hr	Atmo 2D	I	L
clt	Total Cloud Fraction	1-hr	Atmo 2D	M	L
clwvi	Condensed Water Path	3-hr	Atmo 2D	I	T
ct	Air Pressure at Cloud Top	3-hr	Atmo 2D	I	
dds	Near-Surface Dewpoint Depression	3-hr	Atmo 2D	I	
dedt	Tendency of Integrated Cloud Water and Ice	monthly	Atmo 2D	M	
dfq	Horizontal Divergence of Water Vapor Flux	monthly	Atmo 2D	M	
drdt	Tendency of Integrated Water Vapor	monthly	Atmo 2D	M	

Variable	Description	Frequency	Realm	Type	Status
ebq	Residual Term of Atmosphere Water Budget	monthly	Atmo 2D	M	
evspsbl	Evaporation	1-hr	Atmo 2D	M	T
evspsblland	Water Evaporation from Land	3-hr	Land 2D	M	
hfls	Surface Upward Latent Heat Flux	3-hr	Atmo 2D	M	T
hfss	Surface Upward Sensible Heat Flux	3-hr	Atmo 2D	M	L
hrmax	Maximum Near-Surface Relative Humidity	3-hr	Atmo 2D	X	
hrmin	Minimum Near-Surface Relative Humidity	3-hr	Atmo 2D	N	
hurs	Near-Surface Relative Humidity	1-hr	Atmo 2D	I	T
hus	Specific Humidity	3-hr (6-hr)	Atmo 3D	I	L
huss	Near-Surface Specific Humidity	1-hr	Atmo 2D	I	T
lflrt	Lake Floor Temperature	3-hr	Lake 2D	I	
lif	Lake Ice Fraction	3-hr	Lake 2D	I	
lit	Lake Ice Thickness	3-hr	Lake 2D	I	
lmlt	Lake Mixed-Layer Temperature	3-hr	Lake 2D	I	
lmlthick	Lake Mixed-Layer Thickness	3-hr	Lake 2D	I	
mrfsl	Soil Layer Frozen Water Content	monthly	Land 3D	I	
mrfso	Soil Frozen Water Content	3-hr	Land 2D	I	L
mrlsl	Water Content of Soil Layer	monthly	Land 3D	I	

Variable	Description	Frequency	Realm	Type	Status
mrro	Total Runoff	3-hr	Land 2D	M	T
mrros	Surface Runoff	3-hr	Land 2D	M	T
mrso	Total Soil Moisture Content	3-hr	Land 2D	I	
mrsos	Moisture in Upper Portion of Soil Column	3-hr	Land 2D	I	L
pr	Precipitation Flux	1-hr, daily	Atmo 2D	M	T
prc	Convective Precipitation	1-hr	Atmo 2D	M	L
prdc	Deep Convective Precipitation	monthly	Atmo 2D	M	
prfr	Freezing Rain	1-hr	Atmo 2D	M	T
prlp/prra	Liquid Precipitation/Rainfall Flux	3-hr	Atmo 2D	M	T
prrp	Refrozen Rain	3-hr	Atmo 2D	M	
prsn	Snowfall Flux	3-hr	Atmo 2D	M	T
prw	Water Vapor Path	3-hr	Atmo 2D	I	T
ps	Surface Air Pressure	1-hr	Atmo 2D	I	T
psl	Sea Level Pressure	1-hr	Atmo 2D	I	T
pw	Precipitable Water	3-hr	Atmo 2D	I	
rhsmax	Surface Daily Maximum Relative Humidity	daily	Atmo 2D	X	
rhsmin	Surface Daily Minimum Relative Humidity	daily	Atmo 2D	N	
rlds	Surface Downwelling Longwave Radiation	1-hr	Atmo 2D	M	T

Variable	Description	Frequency	Realm	Type	Status
rls	Net LW Surface Radiation	1-hr	Atmo 2D	M	
rlus	Surface Upwelling Longwave Radiation	1-hr	Atmo 2D	M	L
rlut	TOA Outgoing Longwave Radiation	1-hr	Atmo 2D	M	L
rsaa	Shortwave Radiation Absorbed by Atmosphere	1-hr	Atmo 2D	M	
rsds	Surface Downwelling Shortwave Radiation	1-hr	Atmo 2D	M	T
rsdt	TOA Incident Shortwave Radiation	1-hr	Atmo 2D	M	L
rss	Net SW Surface Radiation	1-hr	Atmo 2D	M	
rsus	Surface Upwelling Shortwave Radiation	1-hr	Atmo 2D	M	L
rsut	TOA Outgoing Shortwave Radiation	1-hr	Atmo 2D	M	L
sfcWindmax	Daily Maximum Near-Surface Wind Speed	daily	Atmo 2D	X	L
sic	Sea Ice Area Fraction	3-hr	Ocean 2D	I	
sit	Sea Ice Thickness	3-hr	Ocean 2D	I	
snc	Snow Area Fraction	3-hr	Land 2D	I	
snd	Snow Depth	3-hr	Land 2D	I	L
snm	Surface Snow Melt	3-hr	Land 2D	M	L
snw	Surface Snow Amount	3-hr	Land 2D	I	T
ta	Air Temperature	3-hr (6-hr)	Atmo 3D	I	L
tas	Near-Surface Air Temperature	1-hr, daily	Atmo 2D	I	T

Variable	Description	Frequency	Realm	Type	Status
tasmax	Daily Maximum Near-Surface Temperature	daily	Atmo 2D	X	T
tasmin	Daily Minimum Near-Surface Temperature	daily	Atmo 2D	N	T
tke	Turbulent Kinetic Energy	monthly	Atmo 2D	I	
ts	Surface Temperature	3-hr	Atmo 2D	I	L
tsl	Temperature of Soil	3-hr (6-hr)	Land 3D	I	L
tsmax	Maximum Near-Surface Temperature	3-hr	Atmo 2D	X	
tsmin	Minimum Near-Surface Temperature	3-hr	Atmo 2D	N	
tso	Sea Surface Temperature	3-hr	Ocean 2D	I	
ttop	Air Temperature at Cloud Top	3-hr	Atmo 2D	I	
ua	Eastward Wind	3-hr (6-hr)	Atmo 3D	I	L
uas	Eastward Near-Surface Wind	1-hr	Atmo 2D	I	T
uqvc	Vertically Integrated East. Comp. of Specific Humidity Flux	3-hr	Atmo 2D	M	
uvmax	Maximum Near-Surface Wind Speed	3-hr	Atmo 2D	X	
va	Northward Wind	3-hr (6-hr)	Atmo 3D	I	L
vas	Northward Near-Surface Wind	1-hr	Atmo 2D	I	T
volmrfsl	Soil Layer Volumetric Frozen Water Content	3-hr	Land 3D	I	
volmrlqsl	Soil Layer Volumetric Liquid Water Content	3-hr	Land 3D	I	
vqvc	Vertically Integrated North Comp. of Specific Humidity Flux	3-hr	Atmo 2D	M	

Variable	Description	Frequency	Realm	Type	Status
wa	Upward air velocity	3-hr (6-hr)	Atmo 3D	I	L
zg	Geopotential Height	3-hr (6-hr)	Atmo 3D	I	L
zmla	Height of Boundary Layer	3-hr	Atmo 2D	I	L